Is cosmic fate an inference or an assumption? A preregistered fragility audit of the heat-death conclusion with DESI DR2, Pantheon+, and *Planck*

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Abstract

The DESI era has produced a wave of claims about the ultimate fate of the Universe. We ask how much of a fate verdict is forced by data and how much is induced by assumptions. Under a pre-registered plan, we combine DESI DR2 BAO, *Planck* 2018 distance priors, and Pantheon+/SH0ES supernovae, propagate posterior samples through a frozen fate classifier, and differentiate the verdict along eight assumption axes. Baseline CPL gives $P(\textsc{decay}) = 99.82\%$ and $P(\textsc{rip}) = 0.18\%$, with a nested-sampling point estimate of 0.165%. Within continuous CPL with no point mass at ΛCDM, one hundred ΛCDM mocks yield heat-death-compatible posterior mass consistent with $U(0, 1)$; a zero-residual realization gives 52/48. The real-data posterior tail lies below all 100 mocks ($\hat{p} = 0.0099$, 95% Monte Carlo upper bound 0.0295). A two-parameter Wilks test gives $p = 0.024$, but tests a different statistic and is not an independent calibration of that tail. Translation into fate is assumption-dominated: a quintessence prior sets $P(\textsc{rip}) \equiv 0$ at a fit cost of $\Delta\chi^2 = 8.9$, while Bayesian evidence favors ΛCDM ($\ln B = -1.76$). Four converged $w(a)$ fits have approximate chain-minimum $\Delta\mathrm{AIC} \in [-0.15, 2.54]$, yet $P(\textsc{rip})$ spans 0.21–41.2%; their coordinate priors induce prior rip mass from 25.8–58.1%. A mean-reverting Gaussian process is retained only as a constructive counterexample because its hierarchical chain failed the registered convergence gate. Finally, the registered constraint horizon is *not reached*: BIN4 transports 3.03 nat from its first bin into $a > 1$, while direct observational support ends at $a = 1$. Current fate probabilities therefore combine a measured finite-redshift signal with an unmeasured extrapolation rule; we release the audit framework and pipeline.

I. INTRODUCTION

The DESI era has revived the oldest question in cosmology. Baryon acoustic oscillation measurements from DESI DR2 [1], combined with cosmic microwave background (CMB) data and any of the current Type Ia supernova compilations, prefer a dark-energy equation of state that was recently below -1 and is now rising: the "thawing" quadrant $w_0 > -1$, $w_a < 0$. The statistical strength of this preference is famously unstable—between 2.25σ and 4.2σ depending on which supernova compilation and which CMB likelihood one adopts [1–6]

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—and a growing literature now translates it into statements about the ultimate fate of the Universe [7, 8].

This paper asks a different question. Rather than "what is the fate?", we ask: *of the fate verdict, how much is forced by the data, and how much is donated by the assumptions?* The question has a distinguished qualitative pedigree—Krauss and Turner [9] argued a quarter-century ago that dark energy severs the link between geometry and destiny, and Kallosh and Linde [10] computed observational doomsday bounds for specific potentials—but it has not, to our knowledge, been given a quantitative, preregistered treatment in which the sensitivity of fate probabilities to every analysis choice is itself the measured quantity.

Our design principle is simple: every conclusion is accompanied by its derivative with respect to the assumptions. We freeze an analysis plan before any real-data fate computation (Sec. II), validate the pipeline against the official DESI DR2 chains (Sec. III), propagate every posterior sample to $a = 10^4$ through a frozen fate classifier and report the baseline verdict (Secs. IV–V), calibrate the classifier on one hundred mock universes in which the truth is exactly ΛCDM (Sec. VI), and then measure the fate verdict's response along eight axes of assumption variation (Sec. VII): the supernova compilation, the SH0ES anchor, the CMB information content, the dataset combination, the prior on the dark-energy sector, the inference statistic, the $w(a)$ parametrization, and the position of the truth itself.

Three results organize the paper. First, under a regular continuous CPL model and prior, when the truth is exactly ΛCDM the direction-class posterior mass is consistent with $U(0, 1)$ across repeated datasets. This is a conditional boundary-calibration result, not a theorem about model averaging or priors that put discrete mass on ΛCDM. Second, in depth: the real data push the Big Rip probability to 0.18% under CPL, below all one hundred null mocks ($\hat{p} = 0.0099$, 95% upper bound 0.0295). The Wilks likelihood-ratio value $p = 0.024$ points in the same qualitative direction but concerns a different null statistic. Third, in translation: four converged parametrizations fit the observed window comparably under an exploratory AIC audit, yet the Big Rip probability moves from 0.2% (CPL) to 41% (binned w); under a quintessence prior it is zero before data arrive. A registered mean-reverting GP illustrates how a future mean function fixes a fate, but its nonconverged hierarchical fit is excluded from quantitative comparisons. The registered KL horizon is not reached; $a = 1$ is instead the boundary between direct observational support and grammar-transported future information (Sec. VII E).

We conclude that the current cosmic-fate literature is reporting, with precision, the echo of its own extrapolation conventions. The constructive output of this paper is the audit framework itself: a preregistered, null-calibrated, assumption-differentiated protocol that separates what the data say (a thawing-quadrant signal at the 2–3.3σ level, robust to every dataset ablation we tried) from what the chosen grammar and its induced prior measure say about $a > 1$.

II. PREREGISTERED DESIGN

The analysis plan—model space, priors, dataset combinations, fate-classification criteria, statistical rules, and pass/fail gates—was frozen under a version-control tag (`prereg-v1`) before any real-data fate analysis, and is reproduced in English translation in Appendix A (the frozen original is archived verbatim in the repository). Deviations are permitted only through an append-only amendment log (Appendix B) recording what changed, why, and *when* relative to data contact. Three amendments occurred: A-001 recalibrated the pipeline-validation benchmark after we established that a compressed-CMB likelihood cannot, by construction, reproduce the full-likelihood exclusion significance (Sec. III); A-002 replaced a mock-recovery criterion that is mathematically unsatisfiable for a truth located on a fate boundary (Sec. VI); A-003 added an analytic branch to the fate criteria after the frozen OTHER-budget audit was triggered by parametrizations with bounded future $w(a)$ (Sec. VII D). In each case the trigger, the outcome-independence of the argument, and the sign-off are logged. A later record, A-004, is not a preregistered amendment: it is a post-hoc structural audit of whether the grammar-specific priors can be matched by importance reweighting, and it returns a global No-Go. One boundary of this discipline should be stated plainly: the data predate the plan. The DESI DR2 constraints were public before the freeze, so this preregistration certifies the *integrity of analysis choices*—criteria, gates, and fragility metrics fixed before any fate computation, with both possible verdicts (data-dominated or assumption-dominated fate probabilities) declared equally publishable in advance—not the novelty of a prediction. The published w_0–w_a constraint enters only as the Gate-1 validation target, never as a claim of this paper; the quantities this paper actually reports (null-calibration outcomes, cross-grammar fate tables, constraint-horizon values) did not exist in the literature at freeze time. We additionally logged falsifiable predictions for the null-calibration campaign—six in all, derived from the probability-integral-transform argument, recorded mid-campaign (after ∼17 of 100 mocks) and before its completion; the zero-residual prediction alone preceded its run (six for six correct; Appendix D). The discipline as a whole is the single-author analogue of the blinding now institutional in survey cosmology [11, 12]: where a collaboration shields itself from confirmation bias by blinding catalogs or data vectors, a single author can only freeze the analysis choices in advance and anchor the freeze externally. In a field where the headline significance moves by 2σ under defensible analysis variations, we regard this discipline, not any individual number, as the paper's primary methodological contribution.

III. DATA, LIKELIHOODS, AND VALIDATION

We use three publicly released datasets: DESI DR2 BAO compressed measurements (13 points over seven redshift bins, $0.295 \leq z \leq 2.33$) [1]; the Pantheon+ Type Ia supernova compilation with full stat+sys covariance, with and without the SH0ES Cepheid anchor [13–15]; and *Planck* 2018 information [16] compressed into the distance priors $(R, \ell_A, \Omega_b h^2)$ of Ref. [17], validated there for CPL-type models. Files, versions, and checksums are registered in the repository manifest (Appendix F).

Two validation results define what the pipeline can claim. First, against the official DESI DR2 $w_0 w_a$CDM chains, our marginals agree to $\leq 0.33\sigma$ in every parameter, with Ω_m and H_0 agreeing to $< 0.1\sigma$ (Fig. 1). Second, the compressed CMB prior carries a known, stable information deficit: our ΛCDM exclusion is 2.25σ ($\Delta\chi^2 = 7.44$) where the full-likelihood value is 2.8σ, and the same $\approx 0.5\sigma$ discount recurs across all three supernova compilations (Sec. VII A), consistent with independent compressed-likelihood analyses [3]. Every significance below inherits the compressed-CMB condition; we price it rather than hide it. One procedural lesson is worth recording: compressed priors must be predicted with the same z_*/r_s conventions used at extraction— mixing conventions shifted ℓ_A by 3σ of pure artifact in an early pipeline version, caught by a known-answer test before any science ran.

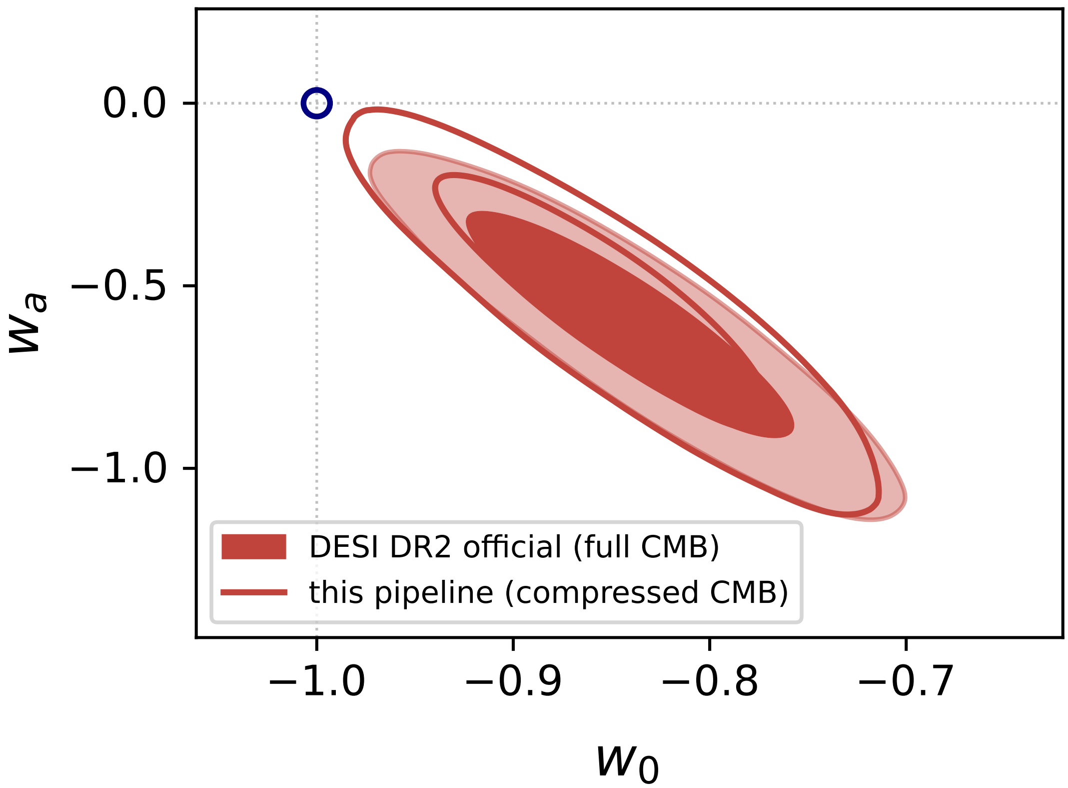

FIG. 1. Gate-1 chain-level validation: this pipeline (compressed CMB, open contours) versus the official DESI DR2 chains (full CMB + lensing, filled). All marginal shifts $\leq 0.33\sigma$. Here and throughout, contour pairs show the 68% and 95% credible regions.

IV. FATE TAXONOMY AND CLASSIFIER

Each posterior sample defines a background $H^2(a)/H_0^2 = \Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_k a^{-2} + \Omega_{\rm DE} f_{\rm DE}(a)$, integrated from $a = 1$ to $a_{\max} = 10^4$ and classified with frozen criteria, in priority order: CRUNCH ($H^2 \leq 0$ reached); RIP (finite-time singularity [18]: convergent $\int da/aH$; for parametrizations whose $w(a)$ has a finite limit w_∞, amendment A-003 substitutes the analytic condition $w_\infty < -1 - \epsilon$); DS (de Sitter asymptotics, $|w + 1| < \epsilon$ with persistent dark energy); DECAY (dark-energy dilution); OTHER (unclassifiable—audited under a frozen 1% budget whose breach triggers the amendment process, as occurred for A-003). Every class carries a boundary flag: samples that flip when thresholds are doubled or halved are reported as such.

The taxonomy is deliberately mapped to thermodynamics: DS *and* DECAY are both heat-

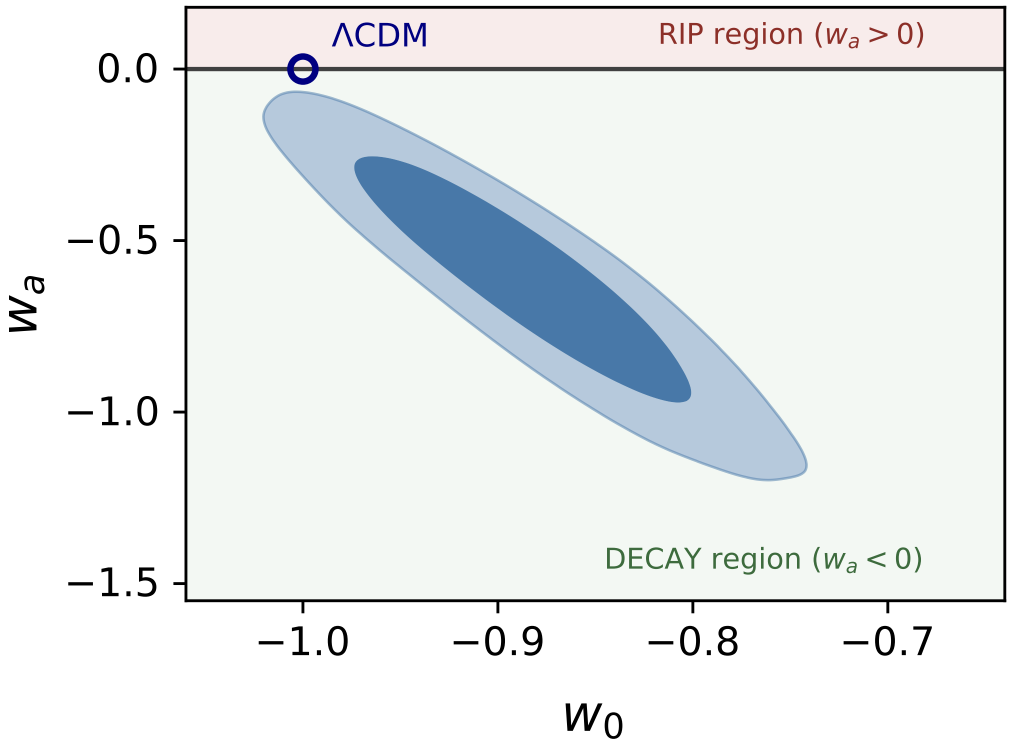

FIG. 2. D0+CPL+P1 posterior over the fate partition. The fate boundary $w_a = 0$ has zero width; ΛCDM sits exactly on it.

death-compatible (eternal expansion)—a point on which parts of the recent fate literature are imprecise. *Decaying dark energy does not rescue the Universe from heat death*; in flat CPL cosmologies the only non-heat-death outcome is the Big Rip. The classifier passes an eight-case unit suite spanning all classes, boundary behavior, and two documented numerical edge cases (Appendix C).

V. BASELINE FATE PROBABILITIES

Figure 2 shows the D0 (DESI DR2 + CMB prior + Pantheon+SH0ES) CPL [19, 20] posterior—$w_0 = -0.884 \pm 0.057$, $w_a = -0.62^{+0.25}_{-0.22}$, $\Omega_m = 0.2997 \pm 0.0050$, $H_0 = 69.00 \pm 0.54$ km s^{-1} Mpc^{-1} (four chains, $R - 1 = 0.0033$)— drawn over the fate partition of the (w_0, w_a) plane. The posterior presses against the zero-width RIP/DECAY boundary at $w_a = 0$; the consequences of that geometry occupy the next section.

Propagating every sample through the classifier yields Table I: $P(\text{DECAY}) = 99.82\% \pm$

TABLE I. Fate probabilities, D0+CPL+P1. *Caution: the direction component of this verdict is uninformative under the null (Sec. VI).*

Class	P	thermodynamic class
DECAY	$99.82\% \pm 0.06\%$	heat-death-compatible
RIP	$0.18\% \pm 0.06\%$ (nested point estimate 0.165%)	non-heat-death
DS	0 (grammar artifact; Sec. VII D)	heat-death-compatible
OTHER/boundary	0 / 0	—

0.06%, $P(\text{RIP}) = 0.18\% \pm 0.06\%$ (batch-means errors), zero OTHER, zero boundary flags. Under the frozen rule that sub-percent tails require a second sampling calculation, nested sampling gives a consistent weighted point estimate, $P(\text{RIP}) = 0.165\%$. The previously quoted 0.038% binomial standard error describes only a fixed-seed equal-weight resample; it is not run-to-run nested-sampling uncertainty and is therefore not used as an error bar.

Two seeds of unease belong here. First, the same data give $\ln B(\text{CPL}/\Lambda\text{CDM}) = -1.76 \pm 0.31$: Bayesian evidence *favors* ΛCDM while $\Delta\chi^2$ disfavors it at 2.25σ—a live Lindley phenomenon [21], developed as a fragility axis in Sec. VII C. Second, $P(\text{DS}) = 0$ exactly: strict de Sitter asymptotics is a measure-zero locus of CPL parameter space, so this zero was decided by the parametrization before any photon arrived. Before interpreting Table I at all, however, one must ask what such a table looks like when the truth is exactly ΛCDM—the subject of the next section.

VI. NULL CALIBRATION

No fate verdict in the literature has, to our knowledge, been calibrated against a null. We generated one hundred mock datasets whose truth is the ΛCDM best fit to D0, drawing noise from the exact likelihood covariances (SN 1657×1657; BAO 13×13; CMB prior 3×3), plus one zero-residual realization for which all three χ^2 vanish at the truth—a closed-loop validation of the mock machinery. Each mock was processed by the full CPL+P1 pipeline.

Because the ΛCDM truth sits on the zero-width CPL fate boundary, the probability integral transform predicts the heat-death-compatible posterior mass to be distributed as $U(0, 1)$ across realizations under the regular continuous model used here: the local likelihood must be calibrated, the boundary locally linear, and the prior absolutely continuous with no point mass on ΛCDM. Under those conditions a direction verdict is correct in $\sim 50\%$ of mocks. This is a property of the specified decision problem, not a statement that pipeline quality is irrelevant and not a universal limit on Bayesian procedures. Spike-and-slab priors, explicit ΛCDM model probability, curved boundaries, or misspecified posterior widths change the result (Appendix E). These predictions were logged mid-campaign (at ∼17 of 100 mocks), before the campaign's completion and before the zero-residual run. Results (Fig. 3): accuracy 50/100 (68% CI $[45, 55]$); KS uniformity $p = 0.25$; mean 0.525 ± 0.030; OTHER and boundary budgets at the 10^{-3} level. (A hundred-realization KS test has limited power against mild distortions, which is why the registered predictions also pinned the mean, the accuracy count, and the budgets—all independently satisfied.) The zero-residual mock returns 52/48. Its data vector equals the truth, but its likelihood covariance remains finite; it is a closed-loop diagnostic, not a zero-uncertainty experiment. The result isolates the role of boundary geometry within this continuous CPL analysis.

The null distribution also supplies the calibrated reading of the real data. Across one hundred ΛCDM universes the lowest Big Rip probability produced by noise is 1.52%; the real Universe sits at 0.18%—eight times deeper, below every realization. The finite-simulation result is $\hat{p} = (0+1)/(100+1) = 0.0099$, with a 95% upper bound of 0.0295. The asymptotic Wilks route gives $p = 0.024$, but the statistics are not equivalent: the mock campaign ranks the one-sided posterior mass $P(w_a > 0 \mid D)$ within a CPL pipeline, whereas Wilks tests the point $(-1, 0)$ against a two-parameter CPL alternative. Their numerical agreement is a useful qualitative cross-check, not two independent measurements of the same tail. The correct summary of Table I is therefore two-layered: *the CPL direction probability has the registered uniform null distribution, while the real data lie unusually deep in its anti-rip tail and also disfavor point-*ΛCDM *by a distinct likelihood-ratio test.* What that signal does and does not imply about the actual fate is the subject of the fragility matrix.

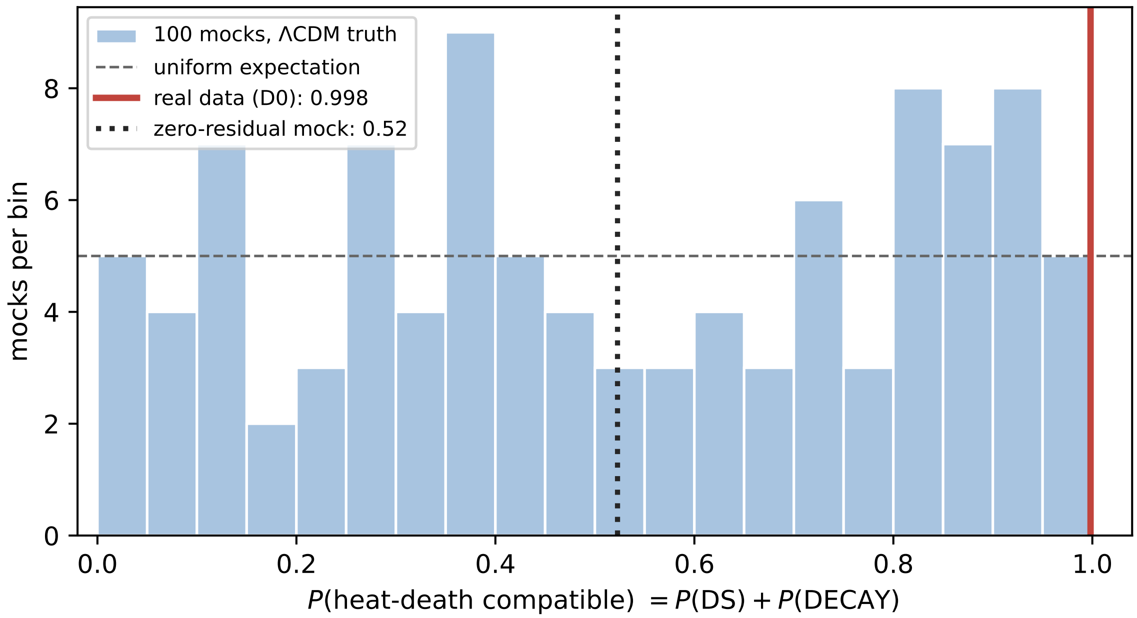

FIG. 3. Null calibration of the fate verdict. Under ΛCDM truth the heat-death-compatible posterior mass is consistent with $U(0,1)$ (KS $p = 0.25$) for continuous CPL+P1; the zero-residual mock returns 52/48. The real-data value (red) lies beyond every mock realization.

VII. THE FRAGILITY MATRIX

A. Data axis

Table II and Fig. 4 show the leave-one-out response. The thawing direction ($w_0 > -1$, $w_a < 0$) survives every ablation—including removal of the CMB (D2; the CMB prior is replaced by a BBN baryon-density prior [22]), which *deepens* rather than erases the preference, a point against the hypothesis that the evidence is purely CMB–low-z tension. The magnitude, by contrast, is fragile: w_a swings threefold (-0.55 to -1.72), and the full combination is more moderate than any probe-removal subset (D2–D4)—the widely quoted $w_a \approx -0.6$ is a negotiated settlement among datasets pulling with different strengths. Swapping the supernova compilation (same model, same priors, same BAO and CMB) moves the ΛCDM exclusion from 2.25σ (Pantheon+) through 2.81σ (DES-Dovekie) to 3.27σ (Union3 [23]), reproducing the literature's significance ladder [4, 5] inside a single pipeline; the SH0ES anchor alone moves H_0 by $+1.35\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ and w_0 by -0.034 —Hubble-tension leakage into the dark-energy sector. Throughout, $P(\textsc{rip})$ stays within 0–0.31%: *fate probabilities*

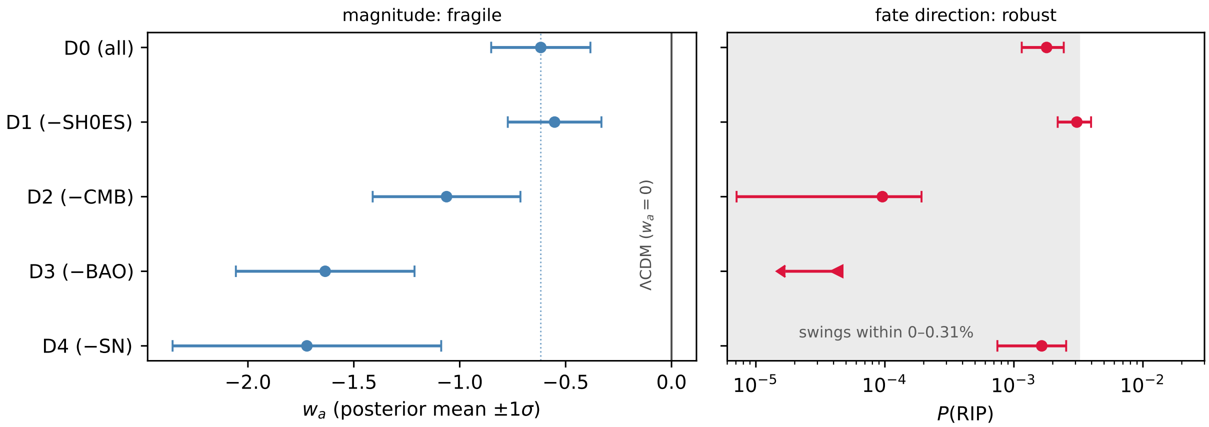

FIG. 4. Leave-one-out forest (CPL+P1 fixed). Left: the w_a magnitude swings threefold across dataset ablations, and the full combination (dotted line) is more moderate than any probe-removal subset (D2–D4). Right: $P(\textsc{rip})$ stays within 0–0.31% throughout (shaded); the D3 entry is the Monte Carlo upper limit for zero RIP samples.

TABLE II. Leave-one-out (CPL+P1 fixed). Per the frozen sub-percent rule, the sub-percent entries were recomputed with nested sampling. The weighted point estimates are consistent with the MCMC values and reproduce the zero tail for D3. Binomial errors stored with the equal-weight resamples are numerical resampling diagnostics, not repeated nested-run uncertainties (per-combination records in the repository).

Combo	removed	w_0	w_a	$P(\textsc{rip})$
D0	—	-0.884 ± 0.057	-0.62	0.18%
D1	SH0ES	-0.848 ± 0.055	-0.55	0.31%
D2	CMB	-0.853 ± 0.067	-1.06	0.01%
D3	BAO	-0.711 ± 0.083	-1.64	0.00%
D4	SN	-0.43 ± 0.22	-1.72	0.16%

are robust to which data one believes (they are not robust to which grammar one believes; Sec. VII D).

B. Prior axis

Under the physical priors the question dissolves before data arrive: P2 (pointwise $w(a) \geq -1$, quintessence) and P3 (P2 + thawing) force $P(\textsc{rip}) = 0$ *identically*—reported, per the frozen plan, as "excluded by prior." The posterior is pinned into a sliver against the $w = -1$ wall ($w_0 = -0.975$, $w_a = -0.015$), and the wall costs the fit $\Delta\chi^2 = +8.9$ relative to P1: a clean measure of the current tension between the data and the quintessence hypothesis. P3 adds nothing to P2 on these data—its extra thawing assumptions are redundant once the pointwise bound binds. The prior axis reading is therefore maximal: $\log_{10} P(\textsc{rip})$ spans from -2.7 to $-\infty$.

C. Inference-statistic axis

The same data, model, and priors yield $\Delta\chi^2 = 7.44$ (Wilks $p = 0.024$, disfavoring ΛCDM) and $\ln B = -1.76 \pm 0.31$ (Jeffreys-scale support *for* ΛCDM). Nothing is inconsistent: the two statistics ask different questions, and the Occam penalty of the 15.5-unit (w_0, w_a) prior area dominates the fit improvement. But the practical consequence is stark—*whether "the evidence favors dynamical dark energy" is true depends on which inference statistic one grants authority*, and the Bayes factor's first-order dependence on prior volume makes it a fragility axis of its own, rarely reported as such. The registered information criteria split along the same fault line: $\Delta\text{AIC} = \Delta\chi^2 + 2\Delta k = -3.4$ sides with the fit, while $\Delta\text{BIC} = \Delta\chi^2 + \Delta k \ln N = +7.4$ sides with the evidence ($\Delta k = 2$, $N = 1673$ data points; negative values favor CPL)—the direction flip is reproduced within the information-criterion family itself.

An explicitly model-averaged reading makes the conditioning visible. As an *exploratory, post-hoc* diagnostic, assign equal prior odds to the two-model set $\{\Lambda\text{CDM}, \text{CPL}\}$. The measured Bayes factor then gives $P(\Lambda\text{CDM} \mid D) = 85.3\%$ and $P(\text{CPL} \mid D) = 14.7\%$; because exact ΛCDM is heat-death-compatible, the corresponding model-averaged compatibility is 99.97%. These values are not privileged—they change with the model set and model prior odds—but they demonstrate why the continuous-CPL 50/50 boundary result cannot be promoted to a universal Bayesian claim.

TABLE III. Exploratory in-window fit audit. Minima are the lowest likelihood χ^2 values in the archived post-burn-in chains; common parameters cancel in ΔAIC. This is not a model-evidence calculation.

Grammar	$\chi^2_{\text{chain,min}}$	$\Delta\chi^2$	ΔAIC	$R-1$
CPL	1489.534	0.000	0.000	0.0077
JBP	1492.075	2.541	2.541	0.0072
BA	1491.523	1.989	1.989	0.0049
BIN4	1485.386	−4.149	−0.149	0.0098

D. Parametrization axis

Table IV and Fig. 5 are the paper's central result, but two comparisons must be separated. First, do the grammars fit the observed window comparably? An exploratory post-hoc audit of the archived chains (Table III) finds approximate minimum likelihood ΔAIC between -0.15 and $+2.54$ relative to CPL. These are chain minima, not dedicated optimizations or evidences, so we claim comparability rather than exact statistical indistinguishability. The literature establishes related in-window degeneracies for overlapping two-parameter families [24], not for this entire set and not for our BIN4 implementation.

Second, the four fits do *not* have identical priors. Their coordinate dimensions, early-matter-domination constraints, and induced measures on functions and fates differ. A deterministic P1 prior audit makes the baseline explicit: prior $P(\text{RIP})$ is 25.8% (CPL), 40.0% (JBP), 58.1% (BA), and 49.7% (BIN4). The likelihood strongly suppresses the rip within every fitted grammar, but the residual posterior remains grammar-dependent. The divergent-future CPL and JBP families deliver near-certain DECAY; bounded-future BA revives a small DS component; and frozen-future BIN4 returns 41.2% RIP, 41.2% DECAY, and 17.6% DS. Its fate-controlling bin is measured as $w_1 = -1.000 \pm 0.044$. A quarter of BIN4 samples carry boundary flags: halving/doubling the DS tolerance moves RIP/DS/DECAY from 45.6/8.7/45.7 to 32.7/34.6/32.7 per cent. The binned reconstruction also relocates the finite-redshift signal: the low-z bin is ΛCDM-consistent and the deviation lies at $z \in [0.7, 1.5]$ ($w_3 = -1.53 \pm 0.20$). Thus CPL's $w_0 \approx -0.88$ is partly the transport of a mid-z signal to $z = 0$ by a linear grammar.

TABLE IV. Grammar-specific induced prior fate mass and posterior fate probabilities (D0 fixed; A-003 criteria). The CPL posterior row uses the generic cross-grammar background machinery and agrees with the baseline within Monte Carlo error. GP* is a construction only.

Grammar	prior $P(\textsc{rip})$	posterior $P(\textsc{rip})$	posterior $P(\textsc{decay})$	posterior $P(\textsc{ds})$	posterior $P(\text{heat-death})$
CPL	25.8%	0.21%	99.79%	0	99.8%
JBP	40.0%	0.79%	99.21%	0	99.2%
BA	58.1%	1.28%	98.35%	0.37%	98.7%
BIN4	49.7%	41.2%	41.2%	17.6%	58.8%
GP*	construction: 0	0	0	100%	100%

A post-hoc matched-prior audit stops before importance weighting. On the declared seven-point $w(a)$ grid, the CPL, JBP, and BA pushforwards occupy different two-dimensional supports and BIN4 occupies a four-dimensional support. Their common intersection contains only constant histories and has probability zero under every continuous native prior. A nondegenerate common target is therefore not dominated by all four pushforwards; diagonal covariance regularization would manufacture overlap rather than identify a matched measure. A-004 accordingly reports a global No-Go and withholds all matched fate probabilities.

The registered Gaussian-process branch is reported differently. Its hierarchical chain stopped with $R-1 = 0.232$, failing Gate 3; moreover the box-truncated hierarchical density was not normalized as a function of its hyperparameters. We therefore remove all GP node, kernel-sensitivity, and fit-comparison claims. One exact construction remains valid without those claims: assigning the conditional-mean future of a stationary GP with mean $w = -1$ makes $w(a \to \infty) = -1$ for every draw, so the asymptotic classifier returns DS by construction. This is a transparent counterexample showing that a future mean function can fix the fate before the likelihood is evaluated; it is not a fifth converged fit. Among the four fitted grammars, $P(\textsc{rip})$ spans 0.21–41.2% and heat-death compatibility 58.8–99.8%. Adding the construction extends the latter endpoint to 100%.

E. Constraint horizon

Where do the data stop speaking? In the binned grammar the posterior-to-prior information $D_{\mathrm{KL}}[p(w(a))\|\pi]$ is 3.03, 2.31, 1.55, and 0.24 nat in the four bins (Fig. 6)—the frozen

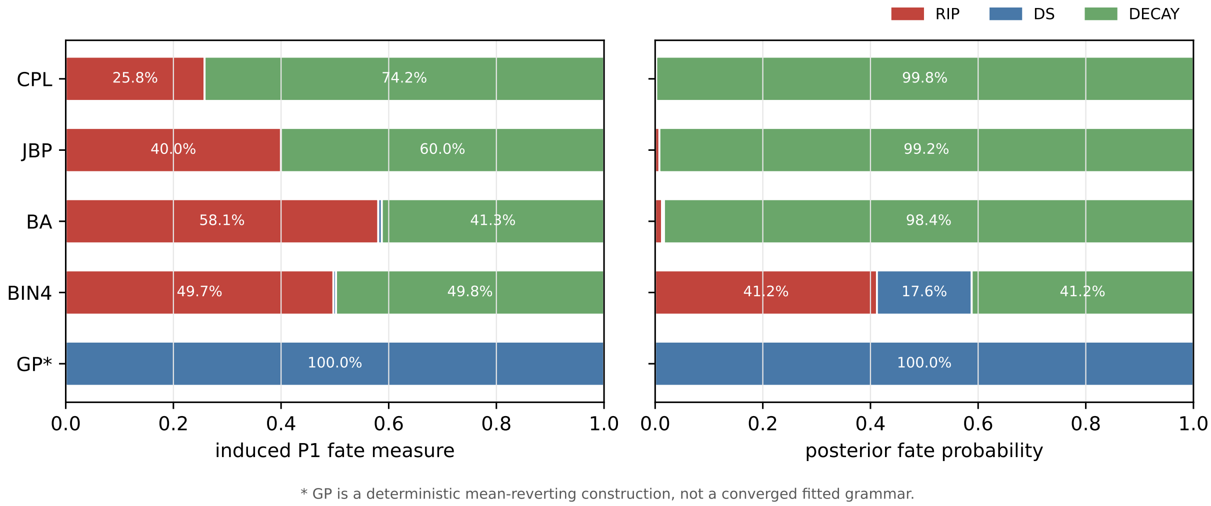

FIG. 5. Prior and posterior fate composition under the grammar-specific P1 measures (A-003 criteria). Flat coordinate priors do not induce a common fate measure. GP* is the deterministic mean-reverting construction, not a converged fitted grammar.

0.1-nat horizon is never crossed inside the observed window: the compressed CMB lever arm constrains the highest-redshift bin weakly but above threshold. The registered metric must then be applied to the registered future rule. BIN4 freezes $w(a > 1) = w_1$, so both its posterior and prior are inherited from w_1 and the future KL remains 3.03 nat. Thus a_h is *not reached*; reporting $a_h = 1$ would be inconsistent with the registered metric.

A separate and scientifically important statement does hold: $a = 1$ is the boundary of *direct observational support*. The nonzero KL plotted for $a > 1$ was learned about w_1 at $a \leq 1$ and transported into the future by the BIN4 freezing rule. CPL transports information differently. Thus future fate is not supported by direct likelihood evaluations at $a > 1$, even though the posterior-to-prior KL of the extrapolated function need not vanish.

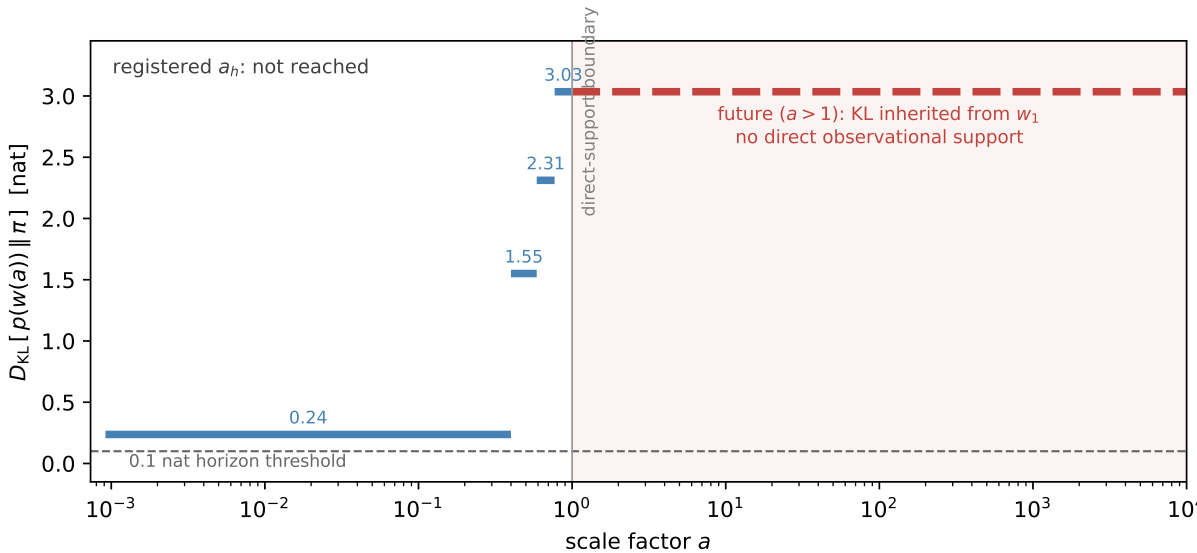

FIG. 6. Constraint horizon (BIN4): posterior-to-prior information in $w(a)$ per bin. The dashed future segment is the w_1 KL inherited by the BIN4 freezing rule, not direct future information. The frozen 0.1-nat threshold is never crossed, so the registered result is a_h: not reached. The vertical line at $a = 1$ marks the distinct boundary of direct observational support.

F. Synthesis

Table V assembles the audit. The stable content is a finite-redshift thawing-quadrant preference: the posterior-tail mock rank and the non-equivalent Wilks statistic both put it near the few-percent level, and its direction survives every data-side variation. Everything specifically eschatological—which fate, with what probability—responds at order unity to theory-side knobs that the current likelihood cannot adjudicate.

VIII. DISCUSSION

Three layers. The audit resolves the fate question into three layers with different epistemic status. (i) *Direction*: whether dark energy will decay or rip has a uniform posterior-mass null distribution within the continuous CPL decision problem; BIN4 places the real posterior near its fate boundary. (ii) *Depth*: the data contain a genuine signal—a thawing-quadrant preference with a posterior-tail mock upper bound of 0.0295 and a distinct Wilks value of 0.024, located at intermediate redshift and robust to every dataset ablation. (iii) *Translation*:

TABLE V. The eight-axis fragility matrix. Each row varies one assumption with all others fixed and reports the response of the fate verdict.

Axis	knob	response	verdict
SN compilation	Pantheon+/Dovekie/Union3	ΛCDM exclusion $2.25-3.27\sigma$	magnitude fragile
SH0ES anchor	in/out	H_0 +1.35 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$; w_0 −0.034	leakage, direction robust
CMB information	compressed/full	stable -0.5σ discount	priced, stable
Dataset ablation	D1–D4	$w_a \in [-1.72, -0.55]$; $P(\textsc{rip})\,0-0.31\%$	direction robust
Prior	P1/P2/P3	$P(\textsc{rip})$: 0.18% → 0 (excluded)	prior decides
Statistic	$\Delta\chi^2$ vs $\ln B$	2.25σ against vs. $\ln B$ for ΛCDM	conditioning flips
Grammar	four fits + GP construction	$P(\textsc{rip})$ 0.21 → 41% in fits	dominant axis
Truth position	ΛCDM in continuous CPL	verdict $\sim U(0,1)$; zero-residual 52/48	conditional null law

converting that signal into eschatology multiplies it by an extrapolation rule and induced prior measure that the data cannot select. The fate literature's headline numbers live in layer (iii).

Why the fragility is structural, not accidental. The fate map is a *discontinuous* functional of parameters that observation constrains only *continuously*: RIP and DECAY are separated by a zero-width boundary, and ΛCDM sits on it. Under regular continuous-model asymptotics, repeated-data class probabilities remain uniform when the truth is exactly on a locally linear boundary, even as the likelihood narrows. This statement is conditional: a discrete ΛCDM component, model averaging, nonregular priors, or a physically derived theory changes the inferential object. In that qualified sense the result is a quantitative instance of the underdetermination of dark energy articulated in Refs. [25, 26], not a no-go theorem for every future Bayesian procedure.

What would change the situation. More precise distance data can narrow the finite-redshift function and may reveal that the truth is off the boundary, but observations made today cannot directly sample $a > 1$. The registered KL horizon is not reached; the relevant limitation is instead the direct-support boundary at $a = 1$. Extrapolation is licensed by trusted theory, not by fit quality alone. CPL and its cousins are more than raw curve fitting—they are compressions calibrated against quintessence-class dynamics inside the window [27–29]—but a calibrated compression is still not a theory: the calibration certifies fidelity to a model class over the fitted range, and extending that fidelity to $a > 1$ is precisely the adoption of the class as a prior—an axis this audit prices (Sec. VII B). A physically derived dark-energy model (a specific potential, a symmetry argument) would convert fate back into a derivation—then data would test the theory inside the window and the theory would carry the license beyond it. Until then, fate probabilities remain conditional predictions of the adopted model class.

An agent-relevant footnote. Our taxonomy classes DS and DECAY as thermodynamically equivalent (both heat death), but they differ for the long-term prospects of computation: a de Sitter horizon imposes a temperature floor that renders total future computation finite [30], while decaying dark energy reopens the Dyson regime in which unbounded subjective experience is compatible with finite free energy [31]. On current data the decay direction is (weakly) preferred—for whatever comfort a 2σ, grammar-conditioned preference is worth.

Relation to the recent fate literature. Papers reporting $P(\text{fate})$ under a single parametrization or physical model [7, 8] are reporting real calculations whose leading uncertainty—the choice of grammar—is absent from their error budgets. Our Table V is the missing error budget. The framework (preregistration, null calibration, differentiation along assumption axes) transfers unchanged to any inference whose conclusion lives beyond its data window.

Limitations. All significances inherit the compressed-CMB condition (priced at $\approx 0.5\sigma$; A-001). That condition is mildest for CPL-like histories, for which the distance priors were validated [17]; for the binned grammar, whose high-z bin can depart further from Λ, extraction-side validity is an additional inherited assumption (the $w_4 < 0$ constraint keeps early dark energy subdominant). The analysis is geometry-only; growth data could shift the window-interior signal. The four fitted grammars are not a model average and their induced priors differ; the prior audit exposes but does not remove that dependence. The A-004 post-hoc audit further shows that this dependence cannot be removed by importance reweighting on the declared seven-point summary, because the grammar pushforwards have incompatible singular supports. The GP is a construction rather than a fit, because the hierarchical chain failed the convergence gate and its truncated hyperprior requires proper normalization before quantitative reuse. The grammar family is not exhaustive—further parametrizations and reconstructions are in active use [24]—so the measured span is not a universal bound. The work was performed by a single author with AI assistance under the verification protocol described; residual unknown-unknowns are bounded only by the audit trail, which is public.

IX. CONCLUSIONS

The data contain a finite-redshift thawing-quadrant preference: the real CPL tail lies below all one hundred ΛCDM mocks, and a distinct Wilks test gives a similar few-percent scale. Translating that preference into fate remains conditional. A quintessence prior excludes the rip before data; four comparably fitting, converged grammars move posterior $P(\text{RIP})$ from 0.21% to 41.2% against induced prior values of 25.8–58.1%; and a mean-reverting GP construction fixes DS without qualifying as a converged fit. The registered a_h is not reached, while $a = 1$ marks the boundary of direct observational support. Current data therefore support a finite-redshift dynamical-dark-energy conditional, not a model-independent cosmic fate. The audit framework makes the conditioning and its error budget explicit.

ACKNOWLEDGMENTS

This work used `CAMB` [32], `cobaya` [33], `GetDist` [34], and `dynesty` [35]; we thank their authors, and the DESI collaboration for the public release of the DR2 cosmology chains used in validation. Substantial portions of the pipeline implementation, literature triage, and manuscript drafting were performed with an AI assistant (Claude, Anthropic) under the preregistered verification protocol described in Sec. II; all analysis choices, amendment sign-offs, and final claims are the author's responsibility.

Appendix A: Frozen analysis plan

The plan below is an English rendering of the document frozen under tag `prereg-v1` (drafted 2026-07-02, tagged 2026-07-03, before any real-data fate analysis); the frozen original is archived in the repository and takes precedence in any discrepancy. Amendments are in Appendix B.

Research question and pre-declared outcomes. The deliverable is a set of fragility measures, not a fate discovery. Two outcome forms were declared in advance as equally publishable: (A) fate probabilities are data-dominated (variation across priors/parametrizations below one order of magnitude), or (B) fate probabilities are assumption-dominated (variation of one order of magnitude or more, or a fate decision falling outside the constraint horizon). The plan records no preference between them.

Model. Background expansion only (no perturbations; growth data out of scope): $H^2(a)/H_0^2 = \Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_k a^{-2} + \Omega_{\rm DE} f_{\rm DE}(a)$, with Ω_r fixed from $T_{\rm CMB} = 2.7255$ K and $N_{\rm eff} = 3.044$ (massless-neutrino approximation, recorded as a limitation) and $\Omega_{\rm DE} = 1 - \Omega_m - \Omega_k - \Omega_r$ by closure. Baseline parametrization CPL, $w(a) = w_0 + w_a(1-a)$; frozen comparison set for the parametrization axis: BA, $w(z) = w_0 + w_a\, z/(1+z^2)$; JBP, $w(z) = w_0 + w_a\, z/(1+z)^2$; BIN4, constant w in four redshift bins $z \in [0, 0.3)$, $[0.3, 0.7)$, $[0.7, 1.5)$, $[1.5, z_{\rm CMB}]$, with the future ($a > 1$) inheriting the first bin; and a Gaussian-process reconstruction, registered as a secondary cross-check (realization and verdict in Sec. VII D).

Priors. P1 (baseline, flat): $\Omega_m \in [0.01, 0.99]$; $H_0 \in [20, 100]$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$; $\Omega_b h^2 \in [0.005, 0.1]$ when the CMB prior is used, else Gaussian 0.02218 ± 0.00055 (BBN, Ref. [22]); $\Omega_k = 0$ (baseline); $w_0 \in [-3, 1]$; $w_a \in [-3, 2]$ with $w_0 + w_a < 0$ (early matter domination); SN absolute magnitude $M_B \in [-20, -18]$, anchored by SH0ES when included, analytically marginalized otherwise. P2 (quintessence): P1 plus the pointwise condition $w(a) \geq -1$ for all $a \in [a_{\rm CMB}, a_{\rm max}]$. P3 (thawing): P2 plus $w_a \leq 0$ and $w_0 \geq -1$ (a phenomenological convention, so labeled).

Data combinations (frozen). D0 = DESI DR2 BAO + CMB distance prior + Pantheon+&SH0ES (baseline); D1 = D0 − SH0ES; D2 = D0 − CMB (BBN prior instead); D3 = D0 − BAO; D4 = D0 − SN; D5 (optional extension) = D0 with the SN compilation swapped.

Fate criteria. As given in Sec. IV, with numerical thresholds frozen in Appendix C, the priority order CRUNCH > RIP > DS > DECAY > OTHER, the $\times 2/\div 2$ boundary flag, the 1% OTHER audit budget whose breach triggers the amendment process, and the frozen thermodynamic mapping (DS, DECAY → heat-death-compatible; RIP, CRUNCH → non-heat-death).

Statistical rules. MCMC via `cobaya`; evidence via nested sampling; convergence $R - 1 < 0.01$ on the worst parameter; nested-model significance via Wilks $\Delta\chi^2$ with explicit p-values; model comparison via ΔAIC, ΔBIC, and $\ln B$; every fate probability reported with batch-means Monte Carlo error and boundary-sample fraction; any tail probability below 1% must be verified by nested sampling before entering the main text.

Fragility measures. $F_{\rm prior}$, $F_{\rm param}$, $F_{\rm data}$: the $\log_{10}$ span of P(RIP) and P(heat-death) across {P1, P2, P3}, {CPL, BA, JBP, BIN4, GP}, and {D0–D4} respectively, each with all other choices fixed; a 100-mock null calibration at the ΛCDM best fit to D0; and the constraint horizon a_h, the scale factor at which the posterior-to-prior KL divergence of $w(a)$ first drops below 0.1 nat (BIN4).

Gates. Gate 1 (pipeline validation, blocking): reproduce the published DESI DR2 w_0w_aCDM result—best fit inside the published 68% contour and ΛCDM-exclusion significance within 0.3σ of the published $\approx 2.8\sigma$ (amended by A-001). Gate 2 (classifier calibration): $\geq 90\%$ of ΛCDM-truth mocks classified heat-death-compatible, and OTHER below 1% (amended by A-002). Gate 3: all chains entering the paper satisfy the convergence rule.

Freeze discipline. After the `git` tag, the plan text is immutable; every change is an append-only amendment entry recording what changed, why, and whether the trigger arose before or after real-data contact; exploratory (unplanned) analyses are permitted but must be labeled as such and kept separate from preregistered results.

Appendix B: Amendment log

The three preregistered-plan amendments are reproduced here in translation; the originals, with sign-off records, are in the repository. A-004 is listed after them as an explicitly post-hoc audit record, not as a fourth preregistered amendment.

A-001 (2026-07-03; during Gate 1, before any fate analysis). **Change:** the Gate-1 significance clause, "within 0.3σ of the literature value ($\approx 2.8\sigma$)," replaced by "consistent with published results using a *compressed* CMB likelihood of the same class" (Ref. [3] reports $N_\sigma \approx 1.1$–2.3 for DESI DR2 + compressed CMB + SN). The best-fit-point clause was unchanged. **Reason:** 2.8σ is a full-*Planck*(+lensing) number; a three-number distance prior discards lensing and spectral-shape information and cannot reach it by construction—an information deficit, not a pipeline error. Evidence: best fit $(w_0, w_a) = (-0.854, -0.520)$ against the official $(-0.838, -0.62)$, inside the 68% contour; CMB-prior $\chi^2 = 1.66$ at the *Planck* ΛCDM mean; our 2.25σ falls inside the published compressed-likelihood range. **Consequence:** every significance in this paper inherits the compressed-CMB condition (priced in Secs. III and VII).

A-002 (2026-07-06; predicted during the mock campaign from the probability-integral-transform argument, confirmed on completion). **Change:** the Gate-2 criterion "$\geq 90\%$ of ΛCDM mocks judged heat-death-compatible" replaced by "the calibration deliverable is the null distribution itself; real-data fate statements must be reported in null-calibrated form.” The OTHER and boundary budgets were unchanged (and were met). **Reason:** the criterion encoded a false factual presupposition. The ΛCDM truth sits on the zero-width RIP/DECAY boundary, so the heat-death-compatible posterior mass is distributed approximately as $U(0,1)$ under the null and *no* calibrated Bayesian procedure can exceed ∼50% direction accuracy (Sec. VI). Measured: 50.0% accuracy, KS $p = 0.25$ against uniformity, zero-residual mock at 52/48, all budgets met, six pre-registered predictions correct—the failure mode matched the mathematical prediction, excluding a pipeline-defect explanation. **Consequence:** Gate 2’s output became a core result of the paper rather than a pass/fail check. The amendment’s “no calibrated procedure” wording is scoped in the revised analysis to the regular continuous CPL decision problem; it does not include priors with a discrete ΛCDM component or model averaging (Sec. VI).

A-003 (2026-07-06; triggered automatically by the frozen 1% OTHER *audit during the first parametrization-axis pass).* **Change:** an asymptotic branch added to the fate criteria: for parametrizations whose $w(a)$ has a finite limit w_∞ as $a \to \infty$, classify analytically—RIP iff $w_\infty < -1 - \epsilon$, DS iff $|w_\infty + 1| \le \epsilon$ with persistent dark energy, DECAY iff $w_\infty > -1 + \epsilon$ ($\epsilon = 0.01$, matching the frozen DS threshold; boundary flagging unchanged). Divergent-w parametrizations (CPL, JBP) keep the original numerical criteria. **Reason:** measured OTHER fractions of 10.5% (BA) and 82.4% (BIN4) breached the budget: bounded or frozen futures fall into a gap between the numerical criteria (decay slower than the DECAY threshold, deviation larger than the DS threshold, power-law rips converging more slowly than the numerical RIP test at $a_{\max}$). The analytic criterion is exact in these cases (a constant $w < -1$ future implies a finite-time singularity). **Consequence for existing results: zero.** CPL and JBP had OTHER $= 0$; D0–D4, P2/P3, and the entire mock calibration are CPL-based and numerically unchanged. Reclassification affected BA and BIN4 only (Table IV).

A-004 (2026-07-21; post-hoc structural audit after all phase-3 results). The candidate common object was the seven-vector $w(a)$ at $a = (0.5, 0.67, 0.8, 1, 1.5, 2, 4)$. CPL, JBP, and BA have two-dimensional pushforward supports in that ambient space; BIN4 has a four-dimensional support. Their common intersection is the one-dimensional set of constant histories, which has probability zero under each continuous native prior. Consequently, no nondegenerate common target is dominated by all four pushforwards and the required importance-density ratios do not exist. An earlier covariance-regularized Gaussian pilot was rejected because it manufactured full-dimensional densities and artificial overlap. The audit is a global No-Go: matched fate probabilities, ESS, and overlap are withheld. It changes no registered analysis or result.

Appendix C: Classifier numerics and known edges

The classifier integrates each posterior sample's background from $a = 1$ to $a_{\max} = 10^4$ and applies Sec. IV's criteria with the frozen thresholds: RIP tail-convergence tolerance 10^{-3} (relative change of $t(a)$ from $a_{\max}$ to $2a_{\max}$); DS tolerance $|w(a_{\max}) + 1| < 0.01$; DECAY thresholds $\rho_{\rm DE}(a_{\max})/\rho_{\rm DE}(1) < 0.01$ or $\rho_{\rm DE}/\rho_m(a_{\max}) < 1$; A-003 asymptotic tolerance $\epsilon = 0.01$. Numerics: the age integral $\int da/(aH)$ is evaluated by adaptive quadrature on 40 logarithmically spaced panels; the CRUNCH scan uses a 400-point logarithmic grid in a; $\ln f_{\rm DE}$ is capped at 700 before exponentiation to avoid overflow (samples beyond the cap are already deep in the RIP regime). The boundary flag re-runs the classification with all thresholds doubled and halved and marks any sample whose label flips.

The implementation passes an eight-case unit suite: ΛCDM → DS (unflagged); phantom $w_a > 0$ → RIP; the DR2 best fit → DECAY; mild thawing → DECAY; a closed, decaying-DE case → CRUNCH; $w_0 = -1.007$, $w_a = 0$ → DS with the boundary flag correctly raised; and two *documented limitation cases* of the frozen criteria, constant-w curves with $w = -0.98$ (quasi-Λ: neither DS-tight nor decayed at $a_{\max}$) and $w = -1.2$ (power-law phantom: a true finite-time singularity whose tail convergence at $a_{\max} = 10^4$ is slower than the frozen tolerance), both of which are intentionally routed to OTHER, i.e. to the manual-audit channel. Constant-w curves are measure-zero in the CPL posterior; the posterior mass in the band $|w_a| < 3\times10^{-4}$ is $\sim 10^{-3}$, safely inside the 1% OTHER budget. These edges were documented before the real-data runs and did not require an amendment; the distinct gap later exposed by bounded-future grammars did (A-003, Appendix B).

For the GP construction the classifier receives $w_\infty = -1$ exactly: the conditional mean of a stationary kernel reverts to its mean function, so the asymptotic branch returns DS for every draw. The registered realization used six nodes at uniform $\ln a$ spacing over $z \in [0, 3]$, a squared-exponential correlation, and mean $w = -1$. Its attempted hierarchical fit multiplied uniform node boxes by $\mathcal{N}(-1, \sigma_f^2 K_\ell)$, with $\sigma_f \in [0.01, 2]$ and $\ell \in [0.2, 3]$. That density was left unnormalized after box truncation, which can change both hyperparameter and node marginals, and the hierarchical chain stopped at $R - 1 = 0.232$. It therefore failed Gate 3 and no quantitative GP posterior is used. The sample-independent asymptotic construction remains valid because it follows from the assigned future mean, not from the nonconverged posterior. Separately, the cross-grammar machinery takes $r_{\rm drag}$ from a ΛCDM computation (drag-epoch physics treated as dark-energy independent); this is self-consistent a posteriori even for the binned grammar, whose free high-z bin sits at $w_4 = -1.71 \pm 0.62$, leaving zero posterior mass with $\rho_{\rm DE}/\rho_m(z_{\rm drag}) > 0.01$.

Appendix D: Mock machinery and zero-residual validation

The null-calibration truth is the ΛCDM best fit to D0 ($\Omega_m = 0.2968$, $H_0 = 68.81$, $M_B = -19.397$), located by repeated bounded minimization. Each of the 100 mocks draws one noise realization per likelihood from the exact covariances used in inference—SN 1657×1657 (stat+sys, after the $z > 0.01$ mask), BAO 13×13, CMB prior 3×3—added to the truth's predicted data vectors, and is then processed by the full CPL+P1 pipeline (MCMC and classifier identical to the real-data runs). The campaign is a point null—mocks are generated at the single best-fit truth rather than marginalized over the ΛCDM posterior. For the direction layer this is sufficient: the uniformity prediction is truth-value-independent for any truth on the fate boundary (Appendix E), and every ΛCDM parameter point sits on it; the depth-layer floor inherits the usual parametric-bootstrap caveat of being conditioned on the fitted truth. Mocks are generated deterministically from a single seed, so the entire campaign regenerates bit-identically from the repository. A 101st, zero-residual realization (`mock000`) provides a closed-loop test: at the truth point all three likelihood χ^2 vanish identically, verifying the mock–likelihood round trip; its fate verdict (52/48) is the finite-covariance boundary diagnostic of Sec. VI.

Six falsifiable predictions were logged mid-campaign (after ∼17 of 100 mocks had been inspected), derived from the probability-integral-transform argument, before the campaign's completion and before the zero-residual realization ran: direction accuracy $50 \pm 10\%$; $P_{\rm heat}$ consistent with $U(0,1)$; 0–3 of 100 mocks below the real data's $P(\textsc{rip})$; OTHER and boundary budgets at the 10^{-3} level; `mock000` near 0.5; and the resulting A-002 trigger. All six were correct. Per-mock records (101 rows), summary statistics, and the batch logs are in the repository.

Appendix E: Boundary calibration statistics

This appendix collects, in an idealized regular-model form, the statistical facts behind the null calibration of Sec. VI: the exact uniformity result, its closed-form verification, the diagnostics that detect a *mis*calibrated pipeline, the finite-mock reporting rules, and the relation to the calibration literature.

One dimension. Let $\hat{x} \sim \mathcal{N}(\theta_0, \sigma^2)$ with posterior $\theta \,|\, \hat{x} \sim \mathcal{N}(\hat{x}, \sigma^2)$ (the calibrated case), and let class A be $\theta > 0$ with the truth on the boundary, $\theta_0 = 0$. Then $P(A \,|\, \hat{x}) = \Phi(\hat{x}/\sigma) = \Phi(Z)$ with $Z \sim \mathcal{N}(0,1)$: by the probability integral transform [36], the class probability is *exactly* $U(0,1)$ across realizations. Numerically (2×10^5 replications): mean 0.5010, variance 0.08356 against the uniform value $1/12 \approx 0.08333$, direction rate 0.502, KS $p = 0.16$. Because the class probability is a continuous functional of the data, its null distribution has no atom at 1/2: individual realizations can look arbitrarily confident while the ensemble verdict remains a fair coin.

Two dimensions, correlated errors. In a (u, v) analogue of $(w_0{+}1, w_a)$ with a D0-like degenerate covariance and a linear fate score $s = \alpha u + v$ ($\alpha = 1.2$, class $s > 0$), the same argument applies to any linear boundary through the truth: 5×10^4 replications give mean 0.5013, variance 0.08326, direction rate 0.505, KS $p = 0.058$ (at 5×10^4 replications the KS test resolves sub-percent distortions; this is comfortable agreement). The CPL fate boundary $w_a = 0$ is exactly linear, so the result covers the geometry of Sec. VI up to the non-Gaussianity of the real posterior—which the full-pipeline campaign tests directly. The derivation assumes a continuous posterior with no atom on the boundary. It does not apply unchanged to spike-and-slab priors or Bayesian model averaging that assigns positive probability to exact ΛCDM.

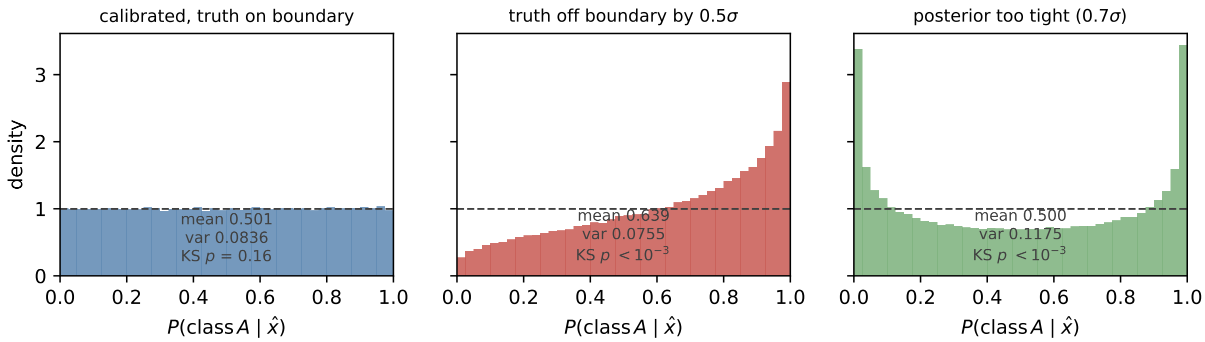

FIG. 7. The boundary-calibration toy (Appendix E). Left: calibrated posterior with the truth on the boundary—class probabilities exactly uniform. Center: truth off the boundary by 0.5σ—direction information appears, as it should. Right: over-confident posterior—mean preserved but variance above 1/12; the uniformity diagnostics fail precisely when the pipeline is defective.

Diagnostics: when uniformity must fail. The value of the null test lies in what breaks it (Fig. 7). Truth off the boundary by 0.5σ: mean 0.639, direction rate 0.692, KS $p \approx 0$—direction information appears exactly when it exists. Decision threshold shifted while the truth sits on it: still uniform (KS $p = 0.25$)—only the truth–boundary distance matters. Posterior too wide (under-confident, 1.5σ): mean stays at 0.500 but the variance collapses to $0.0496 < 1/12$. Posterior too tight (over-confident, 0.7σ): variance inflates to $0.1175 > 1/12$. Strongly curved boundary: KS $p = 2.5 \times 10^{-6}$—uniformity is a locally-linear statement (the CPL boundary is exactly linear). The triple observed in Sec. VI—accuracy $\approx 50\%$, uniform shape, variance consistent with 1/12—is therefore not merely "consistent with" calibration: each defect direction is separately detectable, and none occurred.

Finite-mock reporting rules. With k of n mocks at or below the real-data statistic, we report the plus-one estimate $\hat{p} = (k+1)/(n+1)$ [37] and, for $k = 0$, the 95% upper bound $1 - 0.05^{1/n}$; a Monte Carlo p-value is never reported as zero. These standard rules were adopted during the campaign and applied uniformly; they weaken, not strengthen, the headline bound. For $k = 0$, $n = 100$: $\hat{p} = 0.0099$, upper bound 0.0295—quoted throughout as $p < 0.030$.

Relation to the calibration literature. The lineage runs from the probability integral transform [36] through prequential calibration [38] and posterior predictive checking [39, 40]. The closest modern relative is simulation-based calibration [41], which validates posterior computation via rank uniformity under prior-drawn truths. The present test differs in both conditioning and target: the truth is *fixed at the boundary point*, and the calibrated object is a *decision functional discontinuous there*—the campaign is a null test of the verdict, not of the posterior machinery (which Gate 1 validates separately). The practical residue is a reporting discipline matching the three layers of Sec. VIII: a fate-style claim should state (i) its direction-layer verdict together with the null distribution of that verdict, (ii) its depth-layer tail probability under the finite-simulation rules above, and (iii) the translation-layer conditions—parametrization, prior, statistic—under which the verdict holds; Table V is the translation layer's error budget.

Appendix F: Reproducibility

The repository, `https://github.com/shlinawaf717-collab/cosmo_fate` (archived at Zenodo, `https://doi.org/10.5281/zenodo.21332434`), contains the frozen plan and amendment log, the full pipeline (likelihood configurations, fate classifier and its unit suite, mock generator, batch drivers, nested verification), archived thin chains and per-mock records sufficient for the reported summaries, the figure scripts, and deterministic audits for induced prior fate mass, in-window chain minima, model averaging, the registered constraint-horizon metric, and the A-004 matched-prior structural No-Go. All input data are public: DESI DR2 BAO compressed measurements and covariance; Pantheon+ (with SH0ES), Union3, and DES-Dovekie supernova files obtained via the `cobaya` installer; the CMB distance prior of Ref. [17] (Table I, ΛCDM row, *Planck* 2018 TT,TE,EE+lowE), entered by value; and the official DESI DR2 w_0w_aCDM chains used in Gate 1, downloaded from the public DESI data portal. Every input file is registered in the repository manifest with source, version, and SHA-256 checksum.

Software: Python 3.13, `cobaya` 3.6.2, `camb` 1.6.6, `dynesty` 3.0.0, `getdist` 1.7.7, `numpy` 2.5.0, `scipy` pinned to 1.16.2 (a scipy 1.18 regression in spline evaluation breaks `camb` 1.6.6's BBN interface; the pin and the upstream fix are documented in the manifest). The classifier unit suite, the zero-residual closed-loop test, deterministic mock regeneration, and CI audit invariants together allow every number in this paper to be reproduced from the repository; the environment setup, data installation, and per-result commands are documented in the repository README.

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